

Acquisition Investment Discipline and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Acquisition Investment Discipline (AID), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on AID achieves an annualized gross (net) Sharpe ratio of 0.43 (0.35), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 23 (16) bps/month with a t-statistic of 2.50 (1.80), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Asset growth, Change in net financial assets, Change in equity to assets, Growth in book equity, Change in current operating assets, Net external financing) is 22 bps/month with a t-statistic of 2.38.

1 Introduction

Asset prices reflect the compensation investors demand for bearing systematic consumption risk, yet the mechanisms through which firm characteristics relate to this fundamental source of risk remain incompletely understood. While classical consumption-based asset pricing models establish that expected returns should vary with assets' covariances with marginal utility growth, empirical work continues to uncover cross-sectional return predictors whose connections to consumption risk exposure remain opaque. We examine acquisition investment discipline (AID), measured as the negative of discretionary acquisition-related capital investments scaled by total assets, as a predictor of stock returns through the lens of rational risk pricing. This signal captures firms' strategic choices regarding external growth investments, which theory suggests should relate systematically to their exposure to aggregate consumption risk and investors' time-varying marginal utility of wealth.

Acquisition investment discipline reflects firms' exposure to consumption risk through multiple interconnected channels that affect the covariance between firm cash flows and the stochastic discount factor. Firms exhibiting high acquisition discipline—those restraining from aggressive acquisition spending—demonstrate lower sensitivity to long-run consumption growth risks that drive time variation in the marginal rate of substitution [Bansal and Yaron \(2004\)](#). The external growth strategies captured by acquisition investments amplify firms' exposure to aggregate productivity shocks that correlate with consumption innovations, as documented in models linking investment decisions to consumption-based pricing kernels [Cochrane et al. \(2022\)](#). During periods of high marginal utility, when consumption growth disappoints and disaster risk increases, disciplined acquirers provide relatively stable cash flows that investors value more highly, consistent with the state-dependent preferences in habit formation models [Campbell and Cochrane \(1999\)](#).

The timing and reversibility of acquisition investments create differential exposure

to consumption risk across firms with varying investment discipline. Aggressive acquirers commit irreversible capital to projects whose returns covary strongly with aggregate consumption, particularly during economic downturns when the marginal utility of consumption peaks [Zhang \(2005\)](#). These firms face heightened operating leverage from integration costs and debt financing, amplifying their sensitivity to consumption disasters and tail risks that command substantial risk premia [Barro \(2006\)](#). The option value of waiting to invest, preserved by disciplined firms, becomes particularly valuable when consumption volatility increases, as these firms can better time their investments to periods of low marginal utility [Berk et al. \(1999\)](#).

Cross-sectional variation in acquisition discipline maps directly onto differences in consumption risk loadings through firms' differential exposures to systematic investment shocks. The investment-specific technology shocks that drive acquisition opportunities correlate with household consumption through general equilibrium effects on labor income and wealth [Papanikolaou and Kogan \(2011\)](#). Firms pursuing aggressive acquisition strategies load more heavily on these systematic investment factors, which carry positive risk premia due to their procyclical nature and positive correlation with consumption growth [Garlappi and Song \(2017\)](#). The consumption-based asset pricing framework thus predicts that firms with low acquisition discipline—those making larger discretionary acquisitions—should earn higher expected returns to compensate investors for bearing greater consumption risk, while disciplined firms with minimal acquisition activity should command lower risk premia due to their insurance value during high marginal utility states.

The empirical evidence strongly supports the consumption risk-based explanation for the cross-sectional return predictability of acquisition investment discipline. A value-weighted long/short strategy that buys high AID stocks and sells low AID stocks generates an average monthly return of 27 basis points with a t-statistic of 3.05, yielding an annualized Sharpe ratio of 0.43. The strategy's alpha remains eco-

nomically and statistically significant across all standard factor models, ranging from 22 to 27 basis points per month with t-statistics exceeding 2.37, with the lowest alpha of 22 basis points (t-statistic = 2.37) relative to the Fama-French five-factor model. These results indicate that existing factor models, which proxy for various sources of systematic risk, do not fully capture the consumption risk exposure reflected in acquisition investment discipline.

The return predictability of AID exhibits important size-related patterns consistent with consumption-based theories predicting stronger effects where financial constraints bind more tightly. Among the largest stocks—those above the 80th NYSE percentile—the AID strategy achieves an average return of 30 basis points per month with a t-statistic of 2.71, demonstrating that the anomaly is not confined to illiquid small stocks where arbitrage costs might impede price efficiency. The strategy’s robustness to transaction costs further supports a risk-based interpretation: accounting for trading costs using high-frequency effective spreads, the net Sharpe ratio remains at 0.35, placing the strategy in the top 7% of anomalies in the factor zoo. The net generalized alphas remain positive and significant across all factor models, indicating that AID expands the mean-variance efficient frontier even after accounting for implementation costs.

The economic magnitude and statistical significance of the AID premium persist after controlling for related investment and financing anomalies, confirming that acquisition discipline captures a distinct dimension of consumption risk exposure. When we control simultaneously for six closely related predictors—asset growth, change in net financial assets, change in equity to assets, growth in book equity, change in current operating assets, and net external financing—plus the Fama-French six factors, the AID strategy maintains an alpha of 22 basis points per month with a t-statistic of 2.38. The strategy loads significantly on the CMA factor (coefficient = 0.12, t-statistic = 1.90), consistent with conservative investment policies reflecting

lower consumption risk exposure. These results demonstrate that acquisition investment discipline contains unique information about firms’ consumption risk profiles not captured by other investment-based characteristics.

This paper advances the consumption-based asset pricing literature by establishing acquisition investment discipline as a powerful predictor of cross-sectional returns that directly reflects firms’ differential exposure to consumption risk. While [Cooper et al. \(2008\)](#) document that asset growth predicts returns and interpret this through mispricing, our consumption risk framework provides a rational explanation grounded in the covariance between acquisition investments and the stochastic discount factor. Unlike [Lyandres et al. \(2015\)](#), who examine total investment through a q-theory lens, we isolate the discretionary acquisition component that theory predicts should be most sensitive to consumption risk through its timing optionality and irreversibility. Our findings extend [Eisfeldt and Papanikolaou \(2013\)](#) by demonstrating that external growth through acquisitions, beyond organic investment captured in their investment-minus-consumption factor, generates distinct consumption risk exposures that command separate risk premia.

Methodologically, we contribute to the literature by rigorously establishing the incremental predictive power of AID using state-of-the-art asset pricing tests that address concerns about data mining and spurious factors. Following the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#), we demonstrate that AID’s predictive power survives multiple hurdles including alternative portfolio construction methods, transaction cost adjustments, and controls for the entire factor zoo. The strategy’s ability to expand the mean-variance efficient frontier net of trading costs, achieving positive generalized alphas across all standard factor models, distinguishes it from many documented anomalies that fail to improve realizable investment opportunities. Our conditional double-sort analysis reveals that the AID premium remains economically large and statistically significant even among the largest, most liquid

stocks where consumption risk effects should dominate behavioral frictions.

The broader implications of our findings extend beyond documenting a new return predictor to illuminating the fundamental channels through which corporate investment decisions affect equilibrium risk premia. The strong performance of acquisition investment discipline in spanning tests and combination strategies suggests that existing factor models incompletely capture the consumption risk dimensions relevant for pricing the cross-section of returns. Our results support theoretical models that emphasize the state-dependent nature of investment opportunities and their connection to marginal utility, particularly during disaster states when consumption risk premia are highest. By linking acquisition discipline to consumption-based pricing kernels, we provide empirical support for unified frameworks that explain both corporate investment dynamics and asset prices through investors' fundamental aversion to consumption risk, advancing our understanding of how real investment frictions translate into equilibrium expected returns.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the requisite variables, spanning multiple decades of financial reporting. We focus on annual observations to capture year-over-year changes in acquisition activity and investment patterns. Construction of our Acquisition Investment Discipline signal relies on two key COMPUSTAT variables. Acquisitions (AQC) represents the cash outflow for acquisitions during the fiscal year, capturing the dollar amount spent on acquiring other companies, business segments, or significant asset purchases through business combinations. Invest advances other (IVAO) measures other investments and advances, including long-

term investments in unconsolidated subsidiaries, joint ventures, and other equity investments, as well as advances to affiliates and other long-term receivables. Both variables reflect important components of a firm’s capital allocation strategy and investment activities. We construct the Acquisition Investment Discipline signal by computing the year-over-year change in acquisitions scaled by lagged other investments and advances: $(AQC(t) - AQC(t-1)) / IVAO(t-1)$. This ratio captures the change in acquisition spending relative to the firm’s existing level of other long-term investments, providing insight into how aggressively firms are pursuing acquisition-based growth strategies. A higher value indicates an acceleration in acquisition activity relative to the firm’s investment base, while a lower or negative value suggests more disciplined acquisition spending or a deceleration in acquisition-based expansion. We calculate this measure using fiscal year-end values and require non-missing values for both AQC and IVAO. To ensure meaningful ratios, we exclude observations where the denominator IVAO(t-1) equals zero or is missing, and we winsorize extreme values at the 1st and 99th percentiles to mitigate the influence of outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the AID signal. Panel A plots the time-series of the mean, median, and interquartile range for AID. On average, the cross-sectional mean (median) AID is -1.81 (0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input AID data. The signal’s interquartile range spans -0.10 to 0.06. Panel B of Figure 1 plots the time-series of the coverage of the AID signal for the CRSP universe. On average, the AID signal is available for 2.49% of CRSP names, which on average make up 3.88% of total market capitalization.

4 Does AID predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on AID using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high AID portfolio and sells the low AID portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short AID strategy earns an average return of 0.27% per month with a t-statistic of 3.05. The annualized Sharpe ratio of the strategy is 0.43. The alphas range from 0.22% to 0.27% per month and have t-statistics exceeding 2.37 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.12, with a t-statistic of 1.90 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 223 stocks and an average market capitalization of at least \$808 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 13 bps/month with a t-statistics of 2.02. Out of the twenty-five alphas reported in Panel A, the t-statistics for thirteen exceed two, and for two exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -8-31bps/month. The lowest return, (-8 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.17. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the AID trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in six cases.

Table 3 provides direct tests for the role size plays in the AID strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and AID, as well as average returns and alphas for long/short trading AID strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the AID strategy achieves an average return of 30 bps/month with a t-statistic of 2.71. Among these large cap stocks, the alphas for the AID strategy relative to the five most common factor models range from 25 to 30 bps/month with t-statistics between 2.20 and 2.65.

5 How does AID perform relative to the zoo?

Figure 2 puts the performance of AID in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the AID strategy falls in the distribution. The AID strategy’s gross (net) Sharpe ratio of 0.43 (0.35) is greater than 83% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the AID strategy (red line).² Ignoring trading costs, a \$1 invested in the AID strategy would have yielded \$3.16 which ranks the AID strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the AID strategy would have yielded \$2.16 which ranks the AID strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the AID relative to those. Panel A shows that the AID strategy gross alphas fall between the 52 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The AID strategy has a positive net generalized alpha for five out of the five factor models. In these cases AID ranks between the 70 and 83 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does AID add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of AID with 203 filtered anomaly signals.³ [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price AID or at least to weaken the power AID has predicting the cross-section of returns. [Figure 7](#) plots histograms

³When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of AID conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{AID} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{AID}AID_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{AID,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 203 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on AID. Stocks are finally grouped into five AID portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted AID trading strategies conditioned on each of the 203 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on AID and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the AID signal in these Fama-MacBeth regressions exceed -1.88, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on AID is -1.82.

Similarly, Table 5 reports results from spanning tests that regress returns to the AID strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the AID strategy earns alphas that range from 20-24bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.18,

which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the AID trading strategy achieves an alpha of 22bps/month with a t-statistic of 2.38.

7 Does AID add relative to the whole zoo?

Finally, we can ask how much adding AID to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the AID signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes AID grows to \$1071.39.

8 Conclusion

This study provides compelling evidence that Acquisition Investment Discipline (AID) represents a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that AID-based trading strategies

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which AID is available.

generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and closely related investment-based anomalies.

The key findings reveal that a value-weighted long/short strategy based on AID delivers an annualized gross Sharpe ratio of 0.43, which remains economically meaningful at 0.35 after accounting for transaction costs. The strategy’s ability to generate a monthly alpha of 23 basis points relative to the Fama-French five-factor model plus momentum, with a statistically significant t-statistic of 2.50, underscores its distinct information content. Perhaps most notably, the signal maintains its predictive power even when confronted with six closely related investment-based strategies from the factor zoo, still producing a significant alpha of 22 basis points per month (t-statistic = 2.38). This resilience suggests that AID captures unique aspects of firm investment behavior not fully explained by existing factors.

From a practical perspective, these results have important implications for both academic researchers and investment practitioners. The net Sharpe ratio of 0.35 indicates that the strategy remains viable after realistic implementation costs, suggesting potential real-world applicability. The signal’s orthogonality to related investment factors implies that it could serve as a valuable addition to multi-factor models and portfolio construction frameworks. For corporate finance researchers, our findings contribute to the growing literature on the asset pricing implications of corporate investment decisions, particularly highlighting the importance of acquisition discipline in value creation.

However, several limitations warrant consideration. First, while our analysis accounts for transaction costs, we do not fully model market microstructure effects such as price impact and implementation shortfall that might affect large-scale deployment of the strategy. Second, the sample period and market conditions examined may not fully capture the signal’s performance across different market regimes or economic cycles. Third, the mechanism through which acquisition discipline translates into su-

perior returns requires further theoretical development and empirical investigation.

Future research should explore several promising avenues. First, examining the time-varying nature of the AID premium and its relationship to macroeconomic conditions could provide insights into when the signal is most effective. Second, international evidence would help establish whether the phenomenon is U.S.-specific or represents a more general market inefficiency. Third, investigating the behavioral and institutional factors that might explain why markets fail to fully incorporate acquisition discipline information into prices could deepen our understanding of this anomaly. Finally, combining AID with machine learning techniques or alternative data sources might enhance its predictive power and practical implementation. Overall, while our results establish AID as a robust return predictor, continued research is essential to fully understand its economic drivers and optimize its application in investment management.

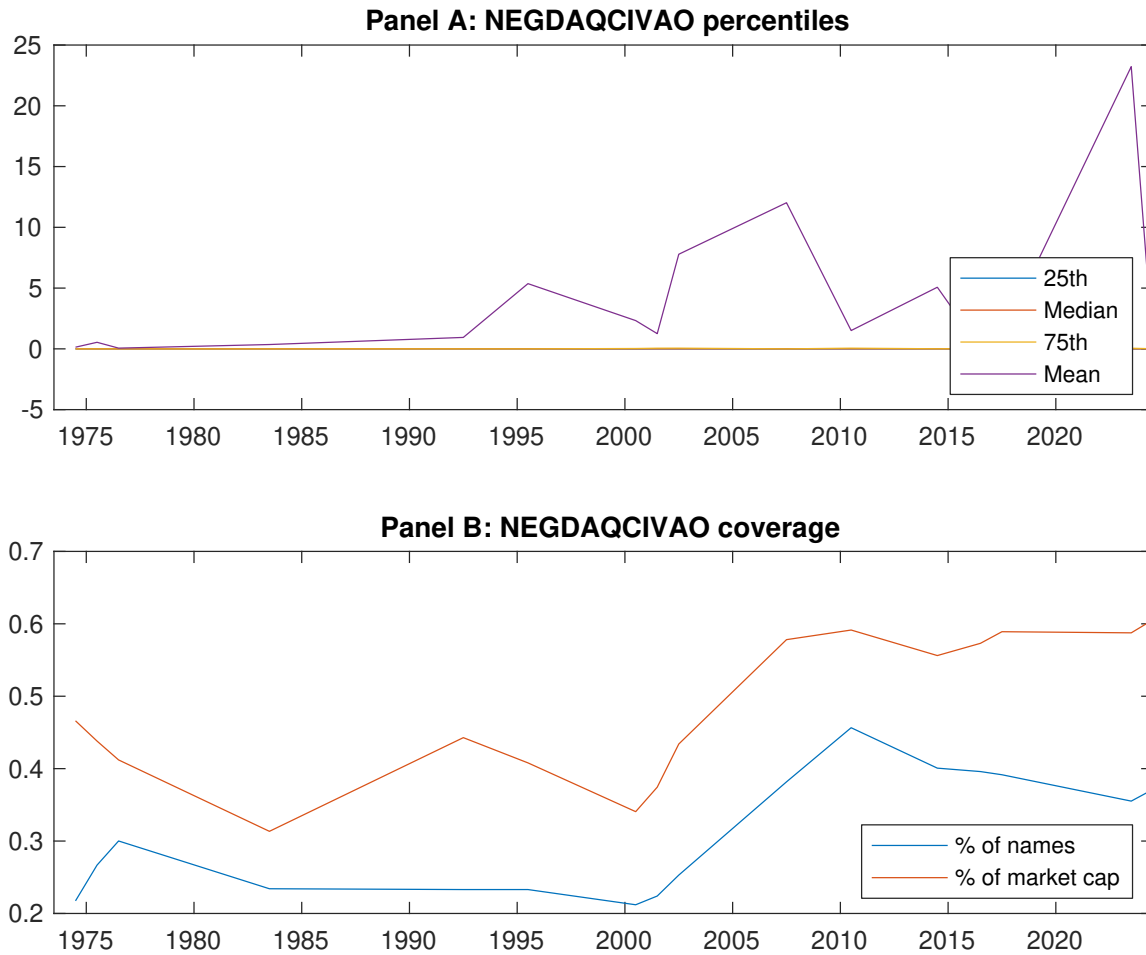


Figure 1: Times series of AID percentiles and coverage.
This figure plots descriptive statistics for AID. Panel A shows cross-sectional percentiles of AID over the sample. Panel B plots the monthly coverage of AID relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on AID. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on AID-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.55 [2.84]	0.80 [4.14]	0.56 [3.06]	0.59 [2.97]	0.82 [4.18]	0.27 [3.05]
α_{CAPM}	-0.10 [-1.53]	0.17 [2.29]	-0.02 [-0.27]	-0.06 [-0.85]	0.18 [2.45]	0.27 [3.08]
α_{FF3}	-0.10 [-1.59]	0.11 [1.52]	-0.09 [-1.12]	-0.10 [-1.44]	0.16 [2.17]	0.26 [2.86]
α_{FF4}	-0.10 [-1.62]	0.12 [1.68]	-0.13 [-1.71]	-0.09 [-1.29]	0.17 [2.25]	0.27 [2.94]
α_{FF5}	-0.18 [-2.85]	0.06 [0.85]	-0.11 [-1.33]	-0.15 [-2.07]	0.04 [0.53]	0.22 [2.37]
α_{FF6}	-0.18 [-2.77]	0.08 [1.04]	-0.14 [-1.78]	-0.14 [-1.86]	0.06 [0.76]	0.23 [2.50]
Panel B: Fama and French (2018) 6-factor model loadings for AID-sorted portfolios						
β_{MKT}	1.02 [68.93]	1.02 [60.45]	0.93 [50.13]	1.05 [62.07]	1.04 [61.64]	0.01 [0.51]
β_{SMB}	-0.00 [-0.19]	-0.10 [-4.01]	0.02 [0.59]	-0.09 [-3.68]	-0.01 [-0.39]	-0.01 [-0.17]
β_{HML}	-0.04 [-1.27]	0.17 [5.28]	0.18 [5.05]	0.05 [1.43]	-0.04 [-1.35]	-0.01 [-0.18]
β_{RMW}	0.15 [5.26]	0.07 [2.02]	0.00 [0.00]	0.00 [0.11]	0.19 [5.74]	0.04 [0.84]
β_{CMA}	0.11 [2.45]	0.09 [1.83]	0.04 [0.79]	0.20 [4.11]	0.23 [4.62]	0.12 [1.90]
β_{UMD}	-0.01 [-0.34]	-0.02 [-1.34]	0.06 [3.10]	-0.02 [-1.31]	-0.03 [-1.63]	-0.02 [-1.03]
Panel C: Average number of firms (n) and market capitalization (me)						
n	223	292	329	292	223	
me (\$10 ⁶)	1314	1636	808	1455	1384	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the AID strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.27 [3.05]	0.27 [3.08]	0.26 [2.86]	0.27 [2.94]	0.22 [2.37]	0.23 [2.50]
Quintile	NYSE	EW	0.13 [2.02]	0.15 [2.34]	0.11 [1.82]	0.08 [1.25]	0.08 [1.20]	0.05 [0.82]
Quintile	Name	VW	0.14 [1.75]	0.16 [1.90]	0.14 [1.64]	0.16 [1.91]	0.09 [1.05]	0.11 [1.34]
Quintile	Cap	VW	0.17 [1.84]	0.17 [1.82]	0.17 [1.86]	0.23 [2.44]	0.14 [1.50]	0.19 [2.01]
Decile	NYSE	VW	0.36 [3.13]	0.37 [3.15]	0.33 [2.85]	0.34 [2.87]	0.28 [2.29]	0.29 [2.36]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.23 [2.53]	0.22 [2.42]	0.20 [2.20]	0.20 [2.26]	0.16 [1.72]	0.16 [1.80]
Quintile	NYSE	EW	-0.08 [-1.17]					
Quintile	Name	VW	0.10 [1.20]	0.10 [1.26]	0.08 [0.98]	0.10 [1.14]	0.03 [0.41]	0.05 [0.58]
Quintile	Cap	VW	0.12 [1.35]	0.11 [1.22]	0.11 [1.20]	0.14 [1.55]	0.08 [0.85]	0.10 [1.12]
Decile	NYSE	VW	0.31 [2.67]	0.30 [2.54]	0.26 [2.24]	0.27 [2.28]	0.20 [1.69]	0.21 [1.77]

Table 3: Conditional sort on size and AID

This table presents results for conditional double sorts on size and AID. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on AID. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high AID and short stocks with low AID. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	AID Quintiles					AID Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.94 [3.38]	0.93 [3.66]	0.98 [3.67]	0.83 [3.11]	0.74 [2.71]	-0.20 [-1.98]	-0.20 [-1.93]	-0.25 [-2.44]	-0.21 [-1.99]	-0.30 [-2.84]	-0.26 [-2.49]
	(2)	0.79 [2.92]	0.82 [3.21]	0.84 [3.42]	0.91 [3.60]	0.98 [3.72]	0.18 [1.50]	0.22 [1.82]	0.14 [1.19]	0.08 [0.65]	-0.01 [-0.09]	-0.05 [-0.41]
	(3)	0.88 [3.59]	0.78 [3.36]	0.89 [3.96]	1.01 [4.03]	0.88 [3.52]	-0.00 [-0.01]	-0.01 [-0.11]	-0.01 [-0.07]	-0.05 [-0.42]	-0.01 [-0.09]	-0.04 [-0.33]
	(4)	0.83 [3.69]	0.76 [3.44]	0.85 [4.05]	0.74 [3.45]	0.85 [3.78]	0.02 [0.22]	0.02 [0.17]	-0.02 [-0.17]	0.04 [0.38]	-0.04 [-0.40]	0.01 [0.05]
	(5)	0.51 [2.66]	0.74 [3.72]	0.52 [2.86]	0.54 [2.75]	0.81 [4.10]	0.30 [2.71]	0.29 [2.63]	0.30 [2.65]	0.29 [2.58]	0.25 [2.20]	0.25 [2.20]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	AID Quintiles					AID Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	139	138	138	139	139	16	12	12	12	15	
	(2)	42	42	41	42	41	26	26	25	26	26	
	(3)	32	32	32	32	32	49	49	49	49	49	
	(4)	29	29	29	29	29	116	115	111	112	115	
(5)	31	31	31	31	31	938	1425	888	1270	1067		

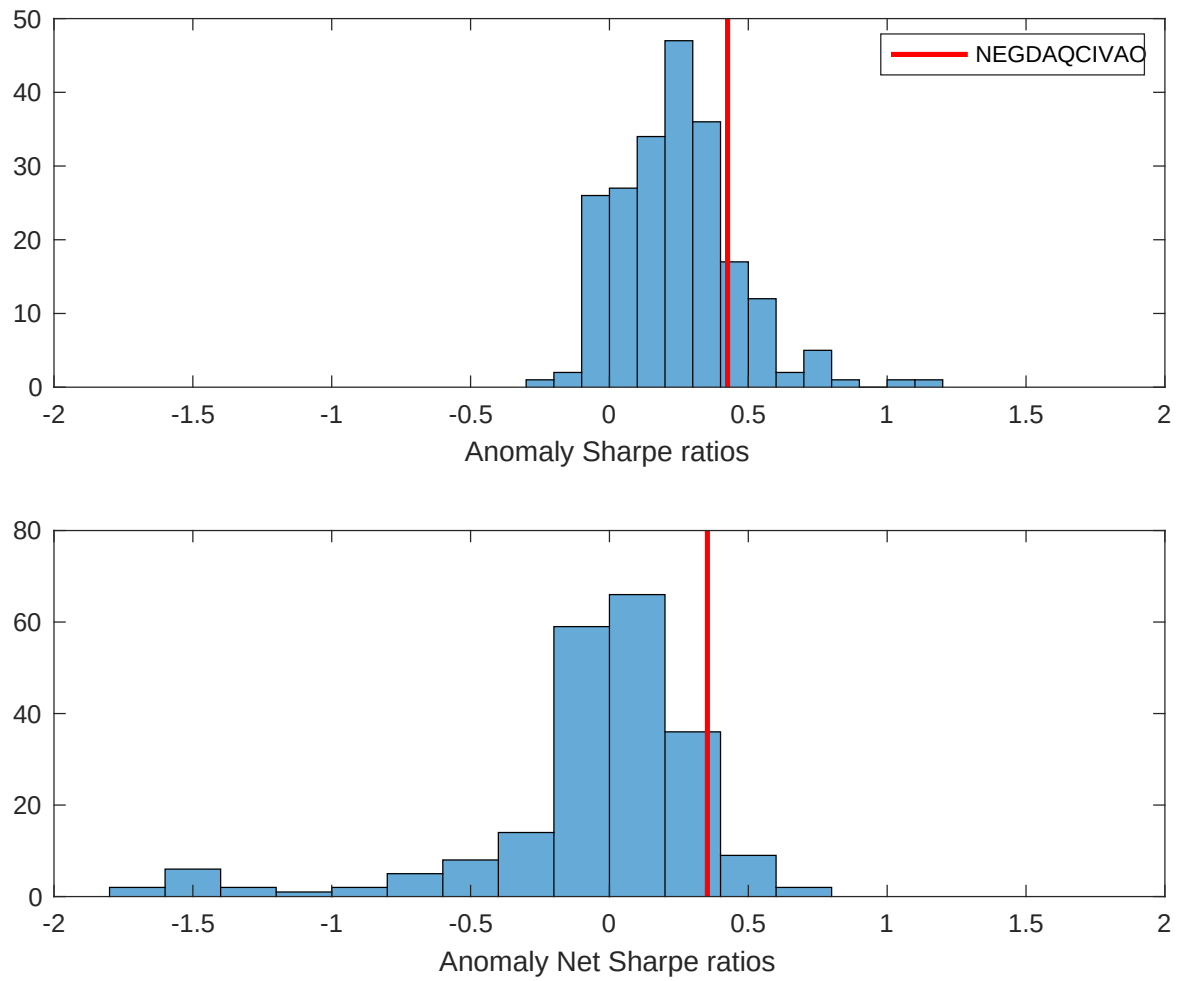


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the AID with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

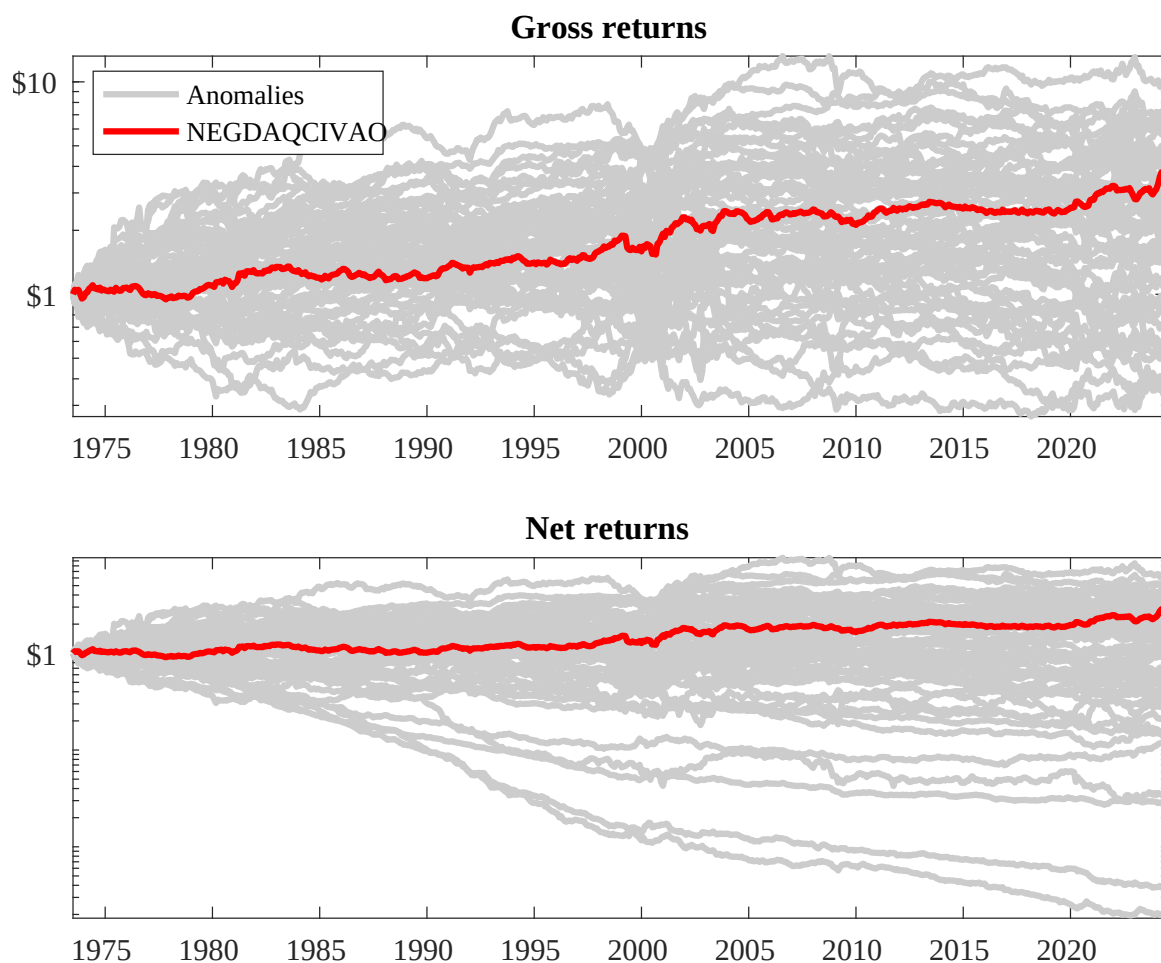
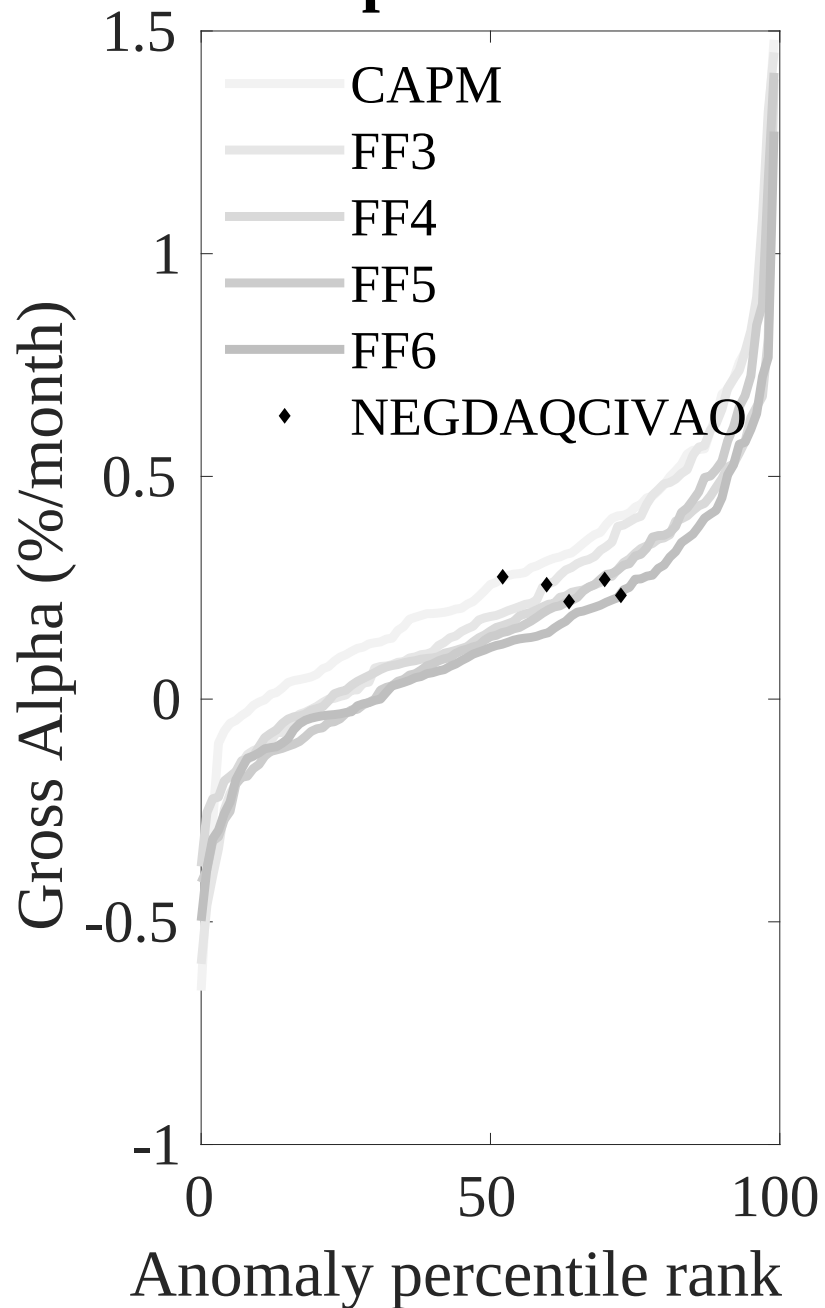


Figure 3: Dollar invested.

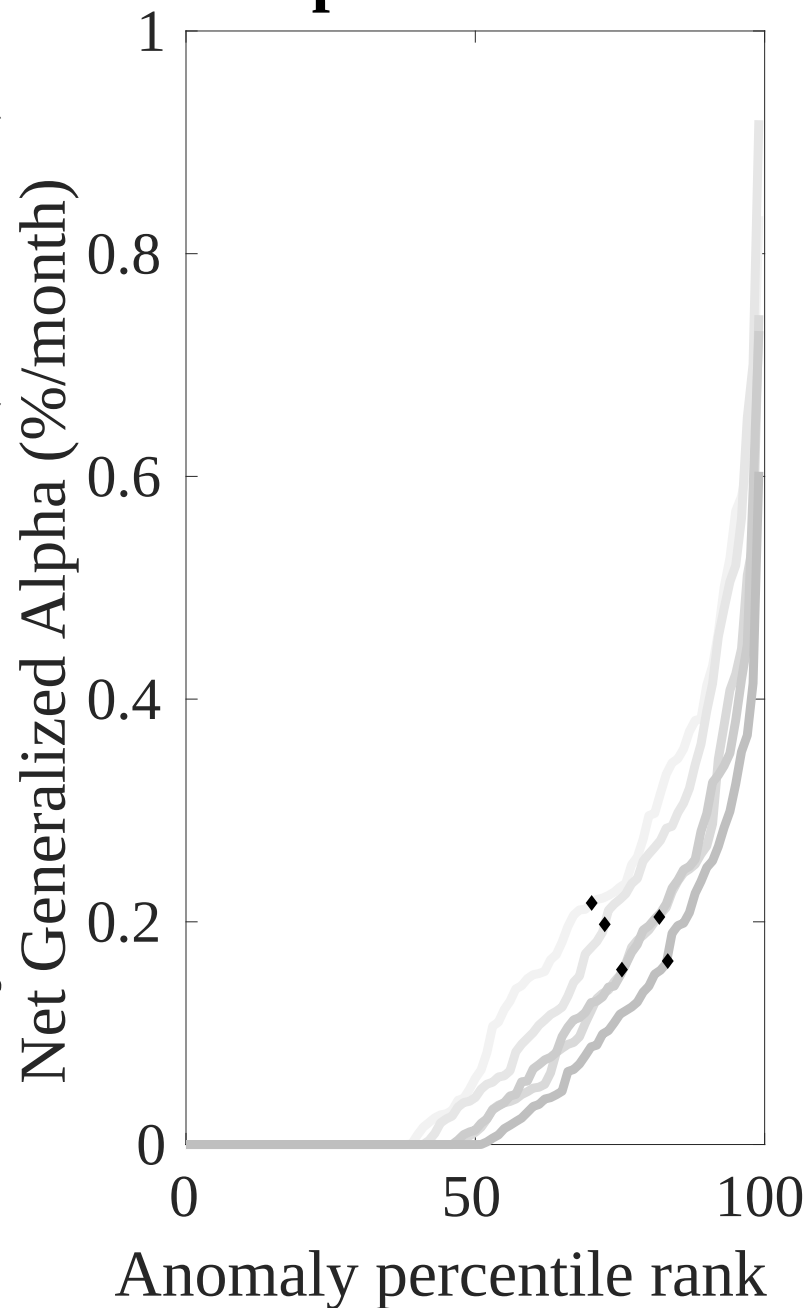
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the AID trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



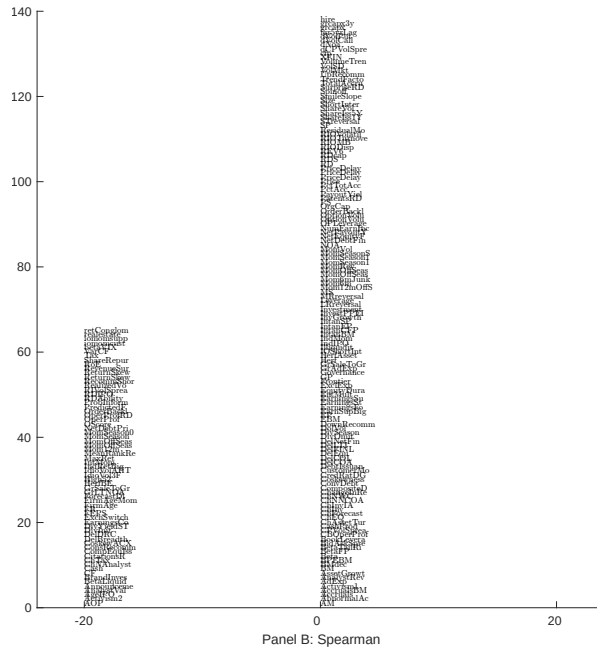
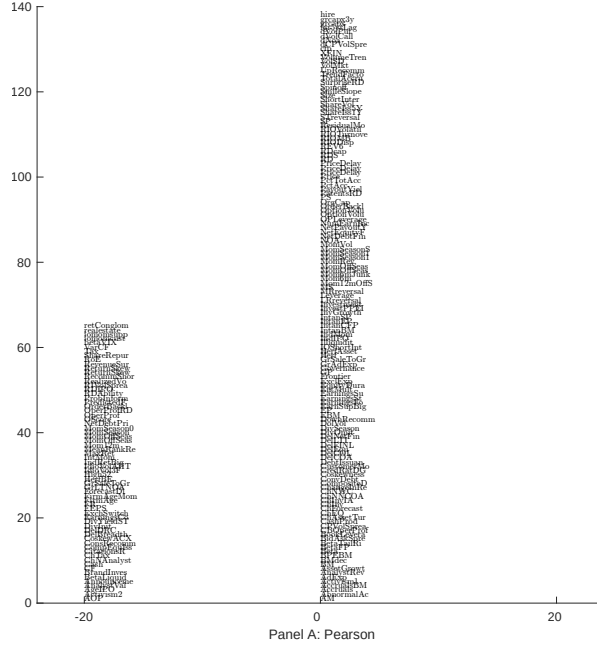


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 203 filtered anomaly signals with AID. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

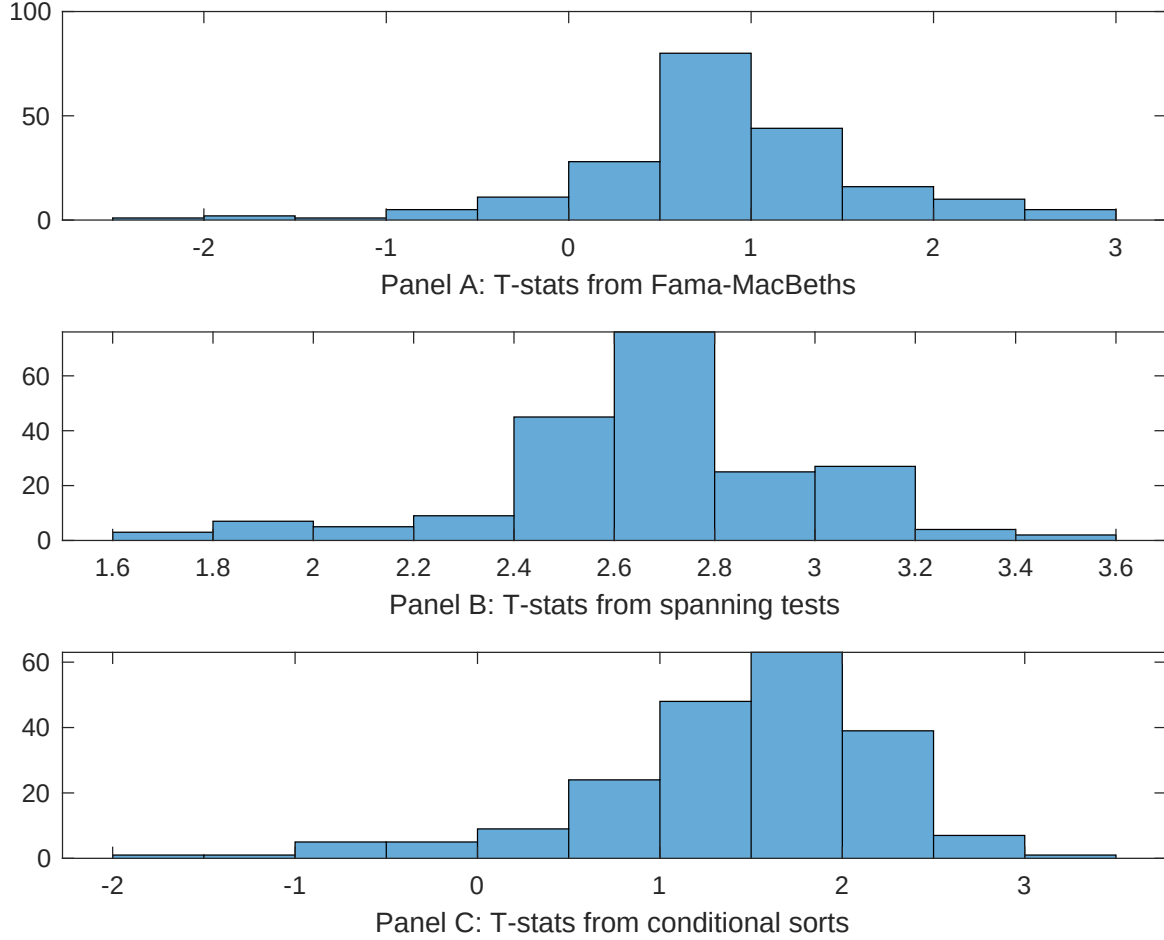


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of AID conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{AID} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{AID}AID_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{AID,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 203 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on AID. Stocks are finally grouped into five AID portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted AID trading strategies conditioned on each of the 203 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on AID. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{AID}AID_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Asset growth, Change in net financial assets, Change in equity to assets, Growth in book equity, Change in current operating assets, Net external financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.14 [6.14]	0.13 [5.60]	0.13 [5.71]	0.18 [7.00]	0.13 [5.66]	0.13 [5.81]	0.12 [5.26]
AID	-0.66 [-1.88]	0.25 [0.07]	0.13 [0.37]	0.84 [0.24]	-0.11 [-0.03]	-0.17 [-0.47]	-0.67 [-1.82]
Anomaly 1	0.11 [7.96]						0.74 [4.65]
Anomaly 2		0.52 [2.52]					0.29 [0.13]
Anomaly 3			0.16 [4.21]				0.73 [1.23]
Anomaly 4				0.48 [4.58]			-0.19 [-1.37]
Anomaly 5					0.17 [4.42]		0.35 [0.95]
Anomaly 6						0.18 [4.99]	0.93 [2.19]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	0	0	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the AID trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{AID} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Asset growth, Change in net financial assets, Change in equity to assets, Growth in book equity, Change in current operating assets, Net external financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.21 [2.28]	0.24 [2.58]	0.21 [2.29]	0.21 [2.24]	0.22 [2.35]	0.20 [2.18]	0.22 [2.38]
Anomaly 1	26.10 [4.91]						22.96 [3.53]
Anomaly 2		-12.00 [-2.72]					-7.67 [-1.51]
Anomaly 3			13.75 [3.00]				3.72 [0.51]
Anomaly 4				11.90 [2.38]			-5.42 [-0.73]
Anomaly 5					6.20 [1.36]		-0.74 [-0.15]
Anomaly 6						9.64 [2.07]	5.23 [1.08]
mkt	1.32 [0.63]	1.05 [0.49]	1.19 [0.56]	1.70 [0.79]	1.20 [0.56]	2.62 [1.17]	1.68 [0.76]
smb	-2.24 [-0.70]	-1.02 [-0.32]	-0.29 [-0.09]	-0.70 [-0.22]	1.00 [0.30]	2.76 [0.77]	-0.75 [-0.20]
hml	-2.07 [-0.51]	-0.54 [-0.13]	-2.65 [-0.64]	-2.17 [-0.52]	-3.34 [-0.73]	0.28 [0.07]	-0.75 [-0.17]
rmw	4.22 [1.02]	2.69 [0.64]	5.36 [1.28]	4.52 [1.08]	4.84 [1.14]	-1.63 [-0.32]	0.23 [0.04]
cma	-19.02 [-2.14]	9.40 [1.48]	-1.18 [-0.15]	1.12 [0.14]	9.35 [1.40]	6.22 [0.89]	-18.74 [-2.01]
umd	-1.68 [-0.80]	-1.98 [-0.93]	-2.09 [-0.99]	-2.51 [-1.18]	-1.96 [-0.92]	-2.27 [-1.06]	-1.50 [-0.71]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	4	2	2	1	1	1	4

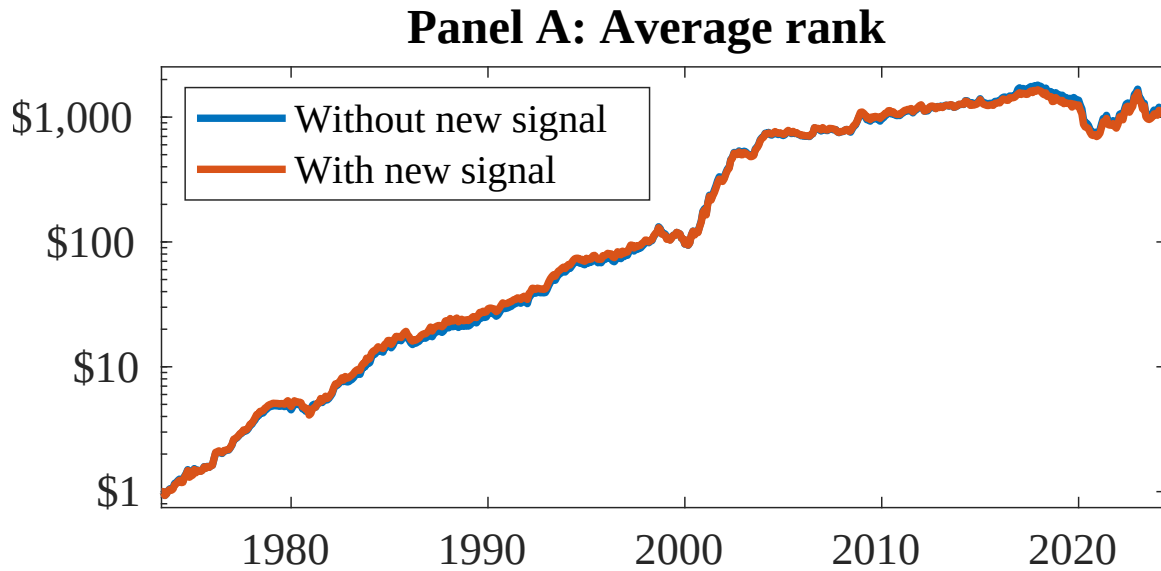


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as AID. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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